

Vol 2 Chapter 2 — The Standard Model Gauge Group, in Operational Detail

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Legacy chapter (May 2026), corrected in place 2026-09-11 (Keeper, K1892) at the sentences the presentation scan flagged; the remaining tiers are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Volume 1 Chapter 8 presented the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ as read off the substrate's primary integers. Two of those readings we no longer make in that form (corrected 2026-09-11): the $SU(3)$ is not supplied by the geometry — the group it carries on the three-dimensional colour slot is $U(1) \cdot SO(3)$, and no compact subgroup of its structure group is $SU(3)$ (Cal Section 946; Spine Lecture 2) — so colour dynamics is *imported* (#108) and the integer $N_c = 3$ is an identification; and electric charge lives in the Cartan of $SO(5)$, not on the $SO(2)$ centre, which is time's rotation (K1687). What the geometry does supply is the electroweak skeleton and the rank-two $SU(2)$. Total Lie-algebra dimension $\dim G_{SM} = 8 + 3 + 1 = 12 = N_c \cdot \text{rank} \cdot 2$.

This chapter is the operational companion: the Weinberg-angle calculation, the gauge-boson mass spectrum, and the substrate-derivations of the gauge couplings that Volume 1 sketched.

2.1 The Weinberg angle

The Weinberg mixing angle θ_W relates the $SU(2)$ and $U(1)_Y$ gauge couplings:

$$\sin^2 \theta_W = \frac{g'^2}{g^2 + g'^2}.$$

Lyra T280 gives the substrate prediction:

$$\sin^2 \theta_W = \frac{N_c}{c_3} = \frac{3}{13} \approx 0.23077.$$

Measured (at M_Z): $\sin^2 \theta_W(M_Z) = 0.23122$. Deviation 0.19% — D-tier (Cal #100 cross-volume tier reconciliation confirmed canonical 0.19% across all volumes).

The two equivalent BST primary forms per Cal Mode 7 algebraic equivalence: Form A N_c/c_3 (Q^5 Chern-class identity); Form B $N_c/(N_c + 2n_C) = 3/13$ (substrate decomposition). Both evaluate to 3/13 at substrate parameters.

The Weinberg angle is, in BST, a ratio of BST primary integers — no tuning. The match at 0.19% to the measured value is one of the framework’s cleanest electroweak predictions.

2.2 The gauge-boson masses

Standard electroweak theory predicts the W and Z boson masses from the Higgs vacuum expectation value v and the gauge couplings:

$$m_W = \frac{1}{2}gv, \quad m_Z = \frac{1}{2}\sqrt{g^2 + g'^2}v = m_W/\cos \theta_W.$$

In BST, the substrate’s natural mass scale enters via $\hbar C_2/c$ (appropriate substrate factor). Lyra’s IP-5 work derives $m_W = 80369$ MeV at 0.5% from substrate primaries combined with the Higgs sector mechanism (Chapter 9). The Z mass follows from the Weinberg-angle relation.

2.3 The Higgs vacuum expectation value

$v = 246.22$ GeV in standard electroweak theory is a fundamental input. In BST, v emerges from substrate Casimir scaling combined with the electroweak symmetry-breaking mechanism of Chapter 9. The substrate-derivation is partial-derived (the full mechanism chain is multi-month research); the value matches at sub-percent precision when the substrate primary combination is identified.

2.4 The coupling constants

$\alpha = 1/N_{\max} = 1/137$ is the substrate’s electromagnetic fine-structure coupling at zeroth order, with the 0.026% residual to experimental $1/\alpha = 137.036$ coming from substrate higher-order corrections (Volume 1 Chapter 10).

The strong coupling $\alpha_s(M_Z) \approx 0.118$ and the weak coupling $g \approx 0.65$ are similarly substrate-derived at the percent level (multi-month refinement ongoing).

2.5 What comes next

Chapter 3 derives the three fermion generations from Q^5 cohomology truncation, with the muon-to-electron mass ratio m_μ/m_e matched at 0.004% (Cal #100 corrected).

Where to look this up: Weinberg angle T280 (Lyra). W mass IP-5. Gauge group derivation Vol 1 Ch 8 + T2436. Standard reference for electroweak: Peskin & Schroeder Chapter 20.